



Welcome to the Else Kooi Laboratory

December 2025 - Standard EKL version



Introduction to the



EKL laboratories

(formerly known as the Dimes Technology Center)

Outline

- 1. Introduction to EKL**
- 2. Training system, SharePoint & NIS**
- 3. Access to the EKL labs, buddy system & start info**
- 4. Cleanroom definition & layout**
- 5. Equipment policy, training & rules**
- 6. Wafer types & wafer handling**
- 7. Ethics & violation of rules**

Introduction to EKL

Else Kooi Laboratory



- EKL takes its name from the Dutch chemist [Else Kooi](#) (1932-2001).
- Main objectives:
 - Facilitate excellent (sub) microfabrication.
 - Support Science & Research as well as Prototyping & Small scale production.
 - Build a bridge between academic research and industrial innovation in the microfabrication field.

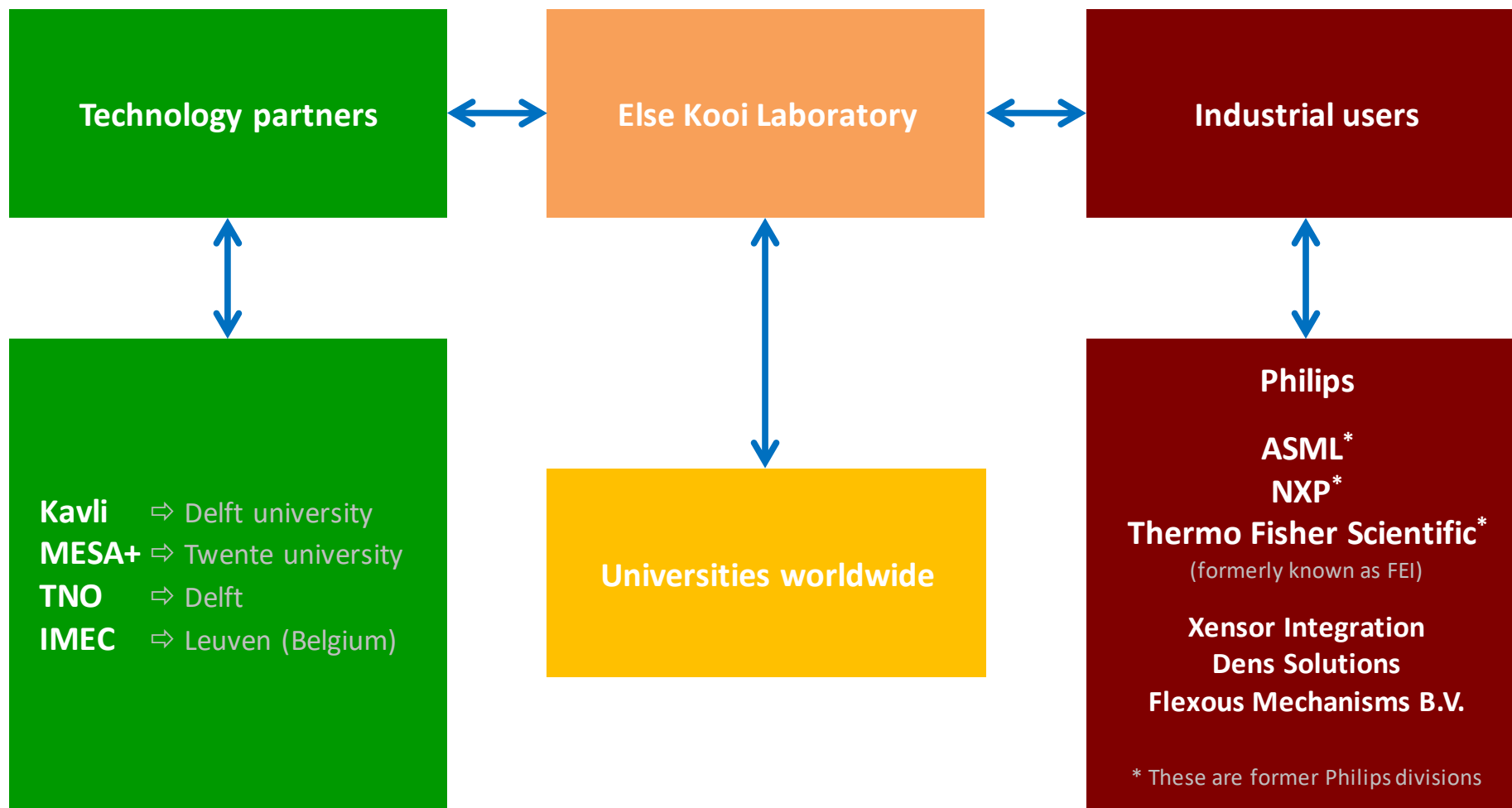


Else Kooi

Else Kooi Laboratory

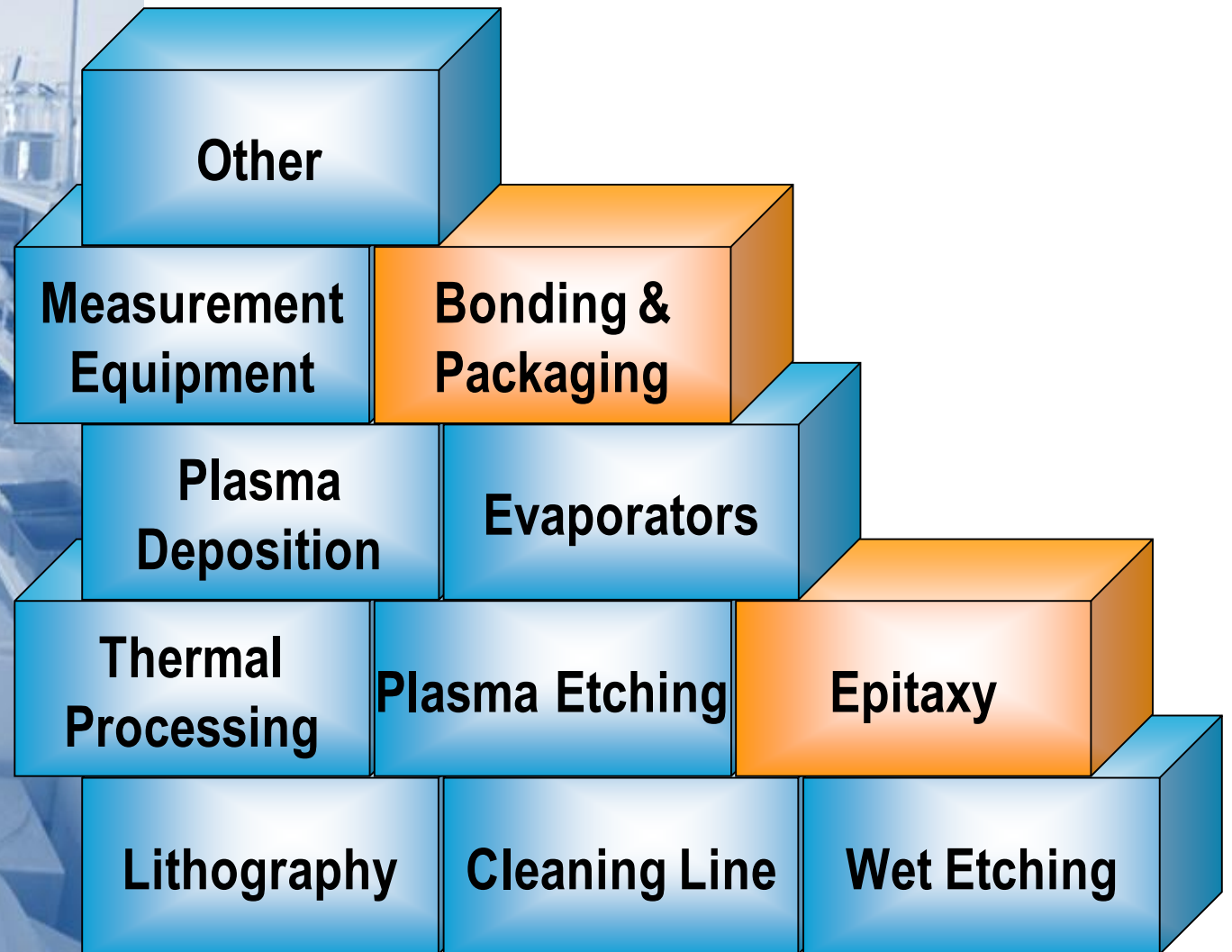
- More than **30 years** of experience.
- More than 150 researchers, PhD students, MSc students , technologists & technicians.
- Processes: Bipolar, BICMOS,
MEMS, TFT, Solar cell,
Polymer technology, Organ on a chip...
- Status of **Preferred Training Center** for many universities and companies world-wide.

Users and partners



EKL Processes

**and
more**



EKL Users profile

- Guest researchers
- Post Doc researchers
- PhD students
- MSc students
- New technical staff
- Semiconductors professionals (e.g. engineers)
- Industrial partners

EKL Users background

- No prior cleanroom experience
- Non-engineering disciplinary background
- Influx from non-electronics industries
- Prior experience at another laboratory or university
- Prior experience with semi-conductor processing

Users from many nationalities

Great cross-cultural interactions

The Netherlands Serbia India Armenia Denmark Bulgaria Hungary Estonia Indonesia China Pakistan
 Syria Australia Turkey Russia France Taiwan Iran Italy Greece Poland Romania Brazil Croatia Cyprus
 Japan Israel Canada Thailand Mexico Ecuador Malaysia Vietnam Singapore Algeria Costa Rica
 Germany Nigeria Switzerland Sweden Spain Iraq Austria Ireland Belgium Egypt Argentina Aruba
 United Kingdom Kazakhstan Portugal Scotland Bolivia Slovakia USA Morocco Sri Lanka Lebanon
 Suriname Ukraine North-Macedonia Philippines Bangladesh Czech Republic Colombia South Korea

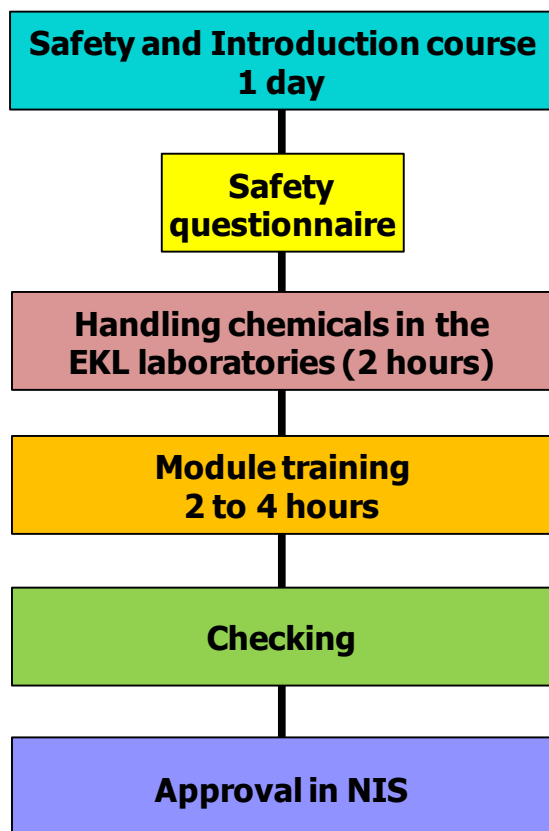


More than **65** nationalities

EKL Training system

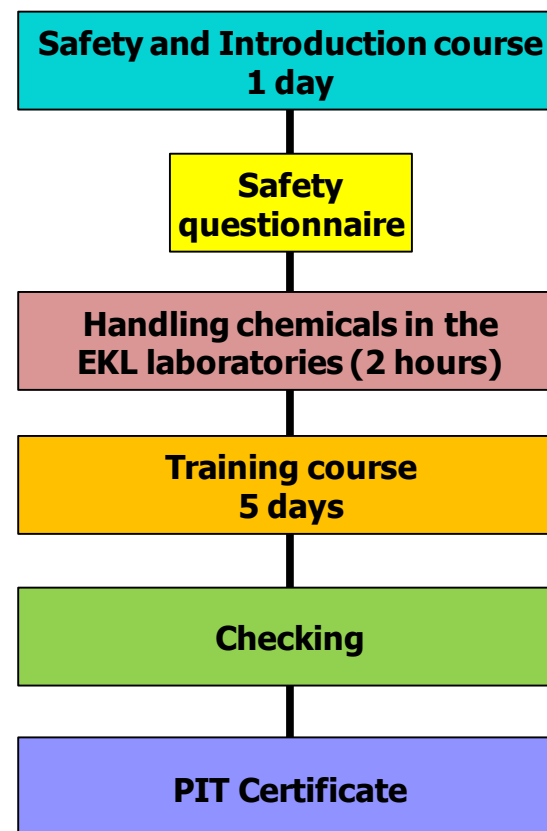
Module Training (MT)

Organized by the tool owner



Process Integration Training (PIT)

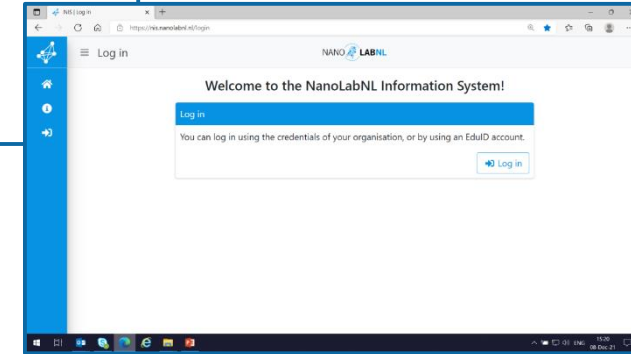
Organized by the training team



SharePoint & NIS

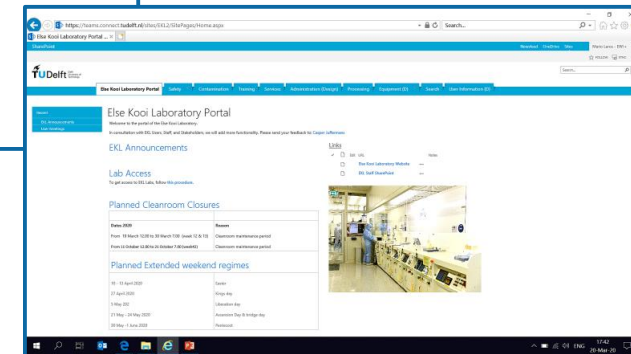
Applying for a **training** must be done through the **NanoLabNL Information System (NIS)**:

<https://nis.nanolabnl.nl>



All **information and documents** for this training can be found on the EKL **SharePoint** website:

<https://sharepoint.tudelft.nl/sites/EKL2>



Training

- A **Module Training** is an equipment, wet bench or fume hood specific training which only includes instructions on how to use that module. Be informed that wet bench training is often required to clean your wafers before other equipment can be used!
- The **Process Integration Training** is a course that includes training on the most used equipment in the cleanroom 100 lab, as well as a hands-on process training that contains useful instructions and tips & tricks on the processing work itself.
- Training requests must be made in the **NanoLabNL Information System**: <https://nis.nanolabnl.nl>
Select: Institute ➡ Courses ➡ Equipment specific courses, and use "Type to search" to find a specific course.
Note: The PIT (Process Integration Training) course can also be found in the list.
- A list of tool owners with their telephone numbers is posted on every tool and can be found in NIS.
Select: Equipment ➡ Search ➡ Type name of tool in search bar ➡ Select tool ➡ Click on Contact.
- Users cannot be trained on a tool by staff members or other users that are **not authorized** to do so.

Access to the EKL labs

The procedure for obtaining access to the EKL labs is described on the [EKL.tudelft.nl](https://www.ekl.tudelft.nl) website under [New Users](#).

Requirements for access:

- ➔ Download the [EKL Access Form](#), fill it in and have it signed by your supervisor. Then submit it to: [Bianca Knot](#) (EKL secretary) or [Vincent van Croonenburg](#) (EKL financial administrator).
- ➔ Acquire an account in the NanoLabNL Information System (<https://nis.nanolabnl.nl>).
- ➔ Attend the EKL [Safety & Introduction](#) course and pass the questionnaire.

When access is granted:

- ➔ Request for activation of your badge from [Bianca Knot](#) or [Vincent van Croonenburg](#).
- ➔ Users must have a **valid badge** in order to enter/exit one of the labs.
 - It is **not allowed** that more than 1 user enters a lab on one badge.
 - Everyone must use their own badge. It is **not allowed** to use the badge of another person.
- ➔ Only **authorized users** may enter the EKL labs.



Refreshment: Once per year a refreshment questionnaire must be passed to keep lab access.

This must be done in the NanoLabNL Information System. You will automatically receive a reminder.

Availability of the EKL labs

- **Standard access:**

- Cleanroom 100, 10.000 and MEMS lab: ⇒ Monday – Friday from 08:00 – 18:00, except holidays

- **Extended access:**

- Cleanroom 100 and 10.000: ⇒ Monday – Friday from 18:00 – 22:00, weekends and holidays
- MEMS lab: ⇒ Monday – Friday from 18:00 – 20:00, weekends and holidays

- **Extended access** is available for experienced users with at least **60 hours** of cleanroom usage.

- Extended access is not given automatically. Your badge needs to be activated for it.
- It can be requested from [Bianca Knot](#) or [Vincent van Croonenburg](#).

- **Access control:**

Access to the labs is controlled by a computer, not a person. This means that you will not be able to open the doors outside your approved access time period, even if this is only 1 second! There is no negotiation possible, so leave the lab in time to prevent getting locked in!

Lab closures

- All labs routinely close down **two weeks** per year (usually planned in April and October) for general **facilities** maintenance. Once a closure are planned, it is fixed and cannot be moved to another date, as external contractors are involved. No work in the lab will be allowed.
- A lab occasionally also has unplanned closures due to a variety of reasons. Closures for emergency reasons cannot be moved. Less urgent closures sometimes can be moved at request. Most of the unplanned closures usually last less than a day. No work in the lab will be allowed.
- When a lab is to be closed, advance notice will be given to all users via NIS by the lab manager ([Bruno Morana](#)), so you can take it into account for your planning. Also notifications will appear on the lab gowning area doors.
- During **public holidays** (e.g. Easter, Christmas) the so-called **weekend regime** applies. During weekend regime periods the lab can still be used, but only by users with extended access!
- For opening hours of the **Office area** check the [Else Kooi Laboratory portal](#) on SharePoint.

EKL Buddy System

- The **buddy system** requires that a **minimum of two** people are inside the EKL labs at all time.
 - A **buddy** is another **authorized and trained** user within the facility, such as a staff member or a good experienced user, who can and will watch over you and care for you in case of trouble.
 - A buddy must have passed the **Handling chemicals in the EKL cleanrooms** course and the **Wet bench Module Training**, and have at least **60 hours of experience** in the EKL labs.
- Any use of **fume hoods** or **cleanroom wet benches** requires the use of the **buddy system**, no matter what time of the day it is.
 - Using equipment that does not make use of wet chemicals does not require the buddy system. (During standard access hours.)



When is a buddy needed?

	DRY PROCESSING	WET PROCESSING	
Involved process equipment	All equipment that does not use wet chemicals.	Pre-filled bathes of the cleanroom 100 wet bench • cleaning line • etching line	Any EKL fume hood or cleanroom 10000 wet bench.
When is a buddy needed?	During: • extended working hours • weekends • public holidays	During: • standard working hours • extended working hours • weekends • public holidays	Always when chemicals are handled. This includes: • transporting • mixing • pouring • disposing
What must be the location of the buddy?	The buddy must be in the EKL building (keep contact over phone) or somewhere in the cleanroom.	The buddy must be in the EKL building (keep contact over phone) or somewhere in the cleanroom.	The buddy must be in the same area as the person that is working with the chemicals.
What is the contact frequency?	Buddies must communicate with each other at least every 15 minutes.	Buddies must communicate with each other at least every 15 minutes.	The buddy must be in sight in the area where the chemicals are being used.

Buddy System rules

- Each lab user is responsible to find his/her own buddy.
- Each lab user is **individually responsible** to ensure that a buddy is **ALWAYS** present during the use of wet benches and fume hoods.
- Lab buddies must contact each other at least every **15 minutes**, to check that all is well.
- A buddy does not continuously have to be the same person and **may change** during lab use.



I NEED A **BUDDY**

Engineer On Duty

- The **Engineer On Duty (EOD)** [Anastasiia Kotliarchuk](#) delivers support in the cleanroom 100 during standard hours.

Assistance can be requested for:

- acquiring:
 - wafers
 - single wafer boxes
 - (extra) manual tweezers
 - cleanroom note books
 - cleanroom paper for a printer
 - batteries
 - help with lost timers
 -
- The engineer on duty can be found in [room EKL00.020](#) or in the lab.
 - When the engineer on duty is not in the lab, wafers can also be acquired from other staff members.

Pregnancy & Health issues

■ Pregnancy

EKL strongly advises pregnant female lab users to inform **their supervisor** and the **EKL safety officer** as soon as possible, as working with certain chemicals can affect the unborn baby.

■ Health issues

It is also advised to inform **your supervisor** and the **EKL safety officer** if you have health issues that can be affected by the use of certain chemicals.

Please note that informing anyone about your condition is voluntarily and not mandatory as this is private information. It will however give us the opportunity to suggest a suitable solution for the work that involves the use of chemicals in any of the labs, to minimize exposure.



Every Cleanroom 100 user will receive

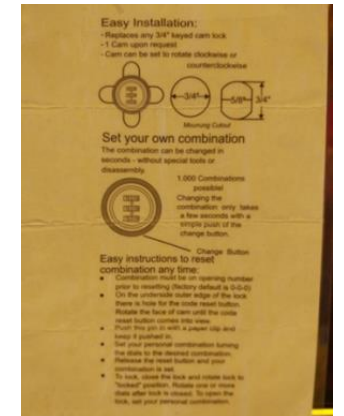
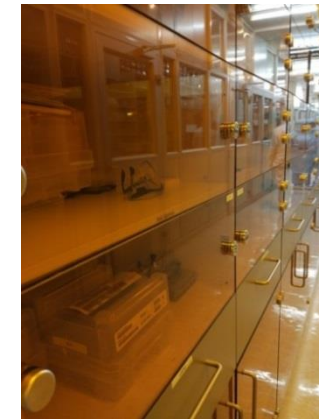
a start package



a CR suit hanger



a cupboard for storage



You can set your own lock combination.

Cleanroom start package

Every CR100 user will receive a start package:

- a CR notebook (**do not use normal paper!**)
- safety glasses
- a manual tweezer
- a timer
- a personal cupboard



PIT participants will receive it at the start of the training.
MT candidates can request it from:

Bianca Knot
or
Anastasiia Kotliarchuk

Don't forget your **student** or **employee** number.

Cleanroom suit hanger



For a cleanroom suit hanger
in the CR100 gowning room:

Make a reservation in NIS
or contact
[Anastasiia Kotliarchuk](#)

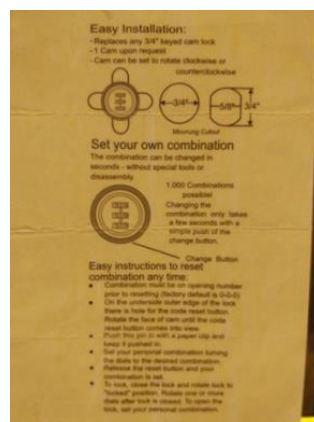


Please exchange
overshoes regularly.



A cleanroom suit should be **changed**
on a **weekly** or **as-needed** base.
(When dirty or torn.)

Cleanroom storage cupboard



You can set your own lock combination.

MSc students will share 1 cupboard.
 PhD students will get max. 1 cupboard.

To obtain a cupboard:

Make a reservation is NIS
 or contact
[Anastasiia Kotliarchuk](#)

After finishing your project, please **empty** the cupboard(s).

Required documents to start a process or experiment

It is **not allowed** to start your process or experiment without an **approved flowchart** with a **run number**. Additionally a **risk assessment** is needed when new chemicals or a new experimental setup will be used.

A **run number** can be obtained by contacting [Hande Aydoğmus](#), our EKL Project Coordinator. You need to include the **first page** of your flowchart. Further instructions will follow.

Note that:

- It is not allowed to make changes to an approved flowchart. A changed flowchart will be regarded as a new flowchart that needs approval.
- A test run also requires an approved flowchart and run number.
- Special permission is required for working with materials that are not C-MOS compatible, to prevent contaminating the lab or equipment.

The first page has to contain **information about your run**:

- Your name, your mentors name, obtained run number, materials and labs that you will use, who approved the flowchart, etc...

A first page template can be found on EKL's SharePoint portal.



The image shows a template for a 'BASIC TRAINING FLOWCHART'. It includes logos for EKL (Else Kooi Laboratory) and TU Delft. The title is 'BASIC TRAINING FLOWCHART' with 'VERSION 01' and 'Valid from July 2018'. Below this is a 'BATCH INFORMATION' table with fields for owner, mentor, run number, wafer amount, mask set, mask box, die size, start date, and PCC status. At the bottom is contact information for the Delft University of Technology Else Kooi Laboratory, including address, phone, fax, and website.

BATCH INFORMATION			
NAME OF OWNER :	Basic Training	MASK SET :	PDM 16
NAME OF MENTOR :	EKL	MASK BOX :	268A and 269 B
RUN NUMBER :	TR0001	DIE SIZE :	10 by 10 mm
WAFER AMOUNT :	6	START DATE :	01 - 01 - 2018
SUBJECT TO PCC	no	PCC APPROVED :	n.a.

**DELFT UNIVERSITY OF TECHNOLOGY
ELSE KOOI LABORATORY**

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P.O. Box : 2011, 2600 GB Delft, The Netherlands
Phone : +31 - (0)15 - 2781668
Fax : +31 - (0)15 - 2622163
Website : www.tudelft.nl/en/onderzoek/facilities/else-kooi-lab

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Risk Assessment

A risk assessment helps us to keep a record of:

- Who might be harmed and how during an experiment.
(by the used chemical / equipment / operation method)
- What you're already doing to minimize the risks.
- What further action is needed to control the risks.
- Who needs to carry out the action.
- When the action will be needed.

Every time a substantial change (e.g. change of chemical or equipment) is done, a new risk assessment **MUST** be made!

Mail to the [EKL Safety Officer](#) to request a copy of the template.

TU Delft		Safety Report Sheet		Version name: EEMCS-052023 Version number: V.D-2022 Page 1 of 8		
Experiment Title: _____						
Report number & Date: _____						
Name of the experiment unit: _____						
User/Author & e-mail: _____						
Section/Research Group: _____						
Location: _____						
Supervisor/Approval: _____						
Area Supervisor: _____						
Review by: _____						
Researcher Phone Number in case of Emergency: _____						
1.1 Experiment Objective: _____						

1.2 Hazardous Materials Used (chemicals, gases, biological agents) & risk groups:						
#	Material Name & formula	CAS#	Chemical state (solid, liquid, gas)	Quantity	Unit	Chemwatch Hazard Ratings*
1						Flammability Toxicity Body contact Reactivity Chronic
2						Flammability Toxicity Body contact Reactivity Chronic
3						Flammability Toxicity Body contact Reactivity Chronic
4						Flammability Toxicity Reactivity Chronic

TU Delft HR.1 Health, Safety and Environment – EEMCS

Wafer boxes



- A **maximum of 3 coloured process boxes** is allowed per cleanroom user. There is no amount limitation for the use of transparent white boxes.
- Always label boxes: **Run ID, your name**, phone number, start date (and comments).
- **Unlabeled boxes will be cleaned up!**
- Return boxes on the shelves for "**dirty boxes**" after your process is finished. They will be cleaned and re-used.

	Run/batch number
	DI 1527
	Operator
	Annie Boddie
Phone	83924
Start date	06-08-2018
Comments:	
Slot 1-6: 6 DSP processwafers	
Slot 7-8: 2 test wafers	

Last person in the lab

The last person in the cleanroom 100 must check if no other persons are left behind or are in trouble, if water taps are closed, hotplates and (microscope) lights are switched off, etcetera, before leaving the lab.

Every person leaving the cleanroom 100 **at 18:00** is considered to be the last person.



Close Di water taps to prevent EKL from running out of water for the next working day.



Switch off hotplates to prevent fire hazards.



Switch off lab lights to save power & prevent microscope bulbs from burning out.

Visitors

- Visitors can enter the laboratories **only after approval** by EKL.
- Visitors and guests must be escorted by a **qualified** staff member or **permitted** user.
- Visitors and guests are admitted to the laboratories for the purposes of a **tour** or **observation only**. Visitors will **not** be allowed to process devices or to use any of the available equipment.
- All visitors will need a **guest badge**. It is not allowed to enter the laboratories without one.

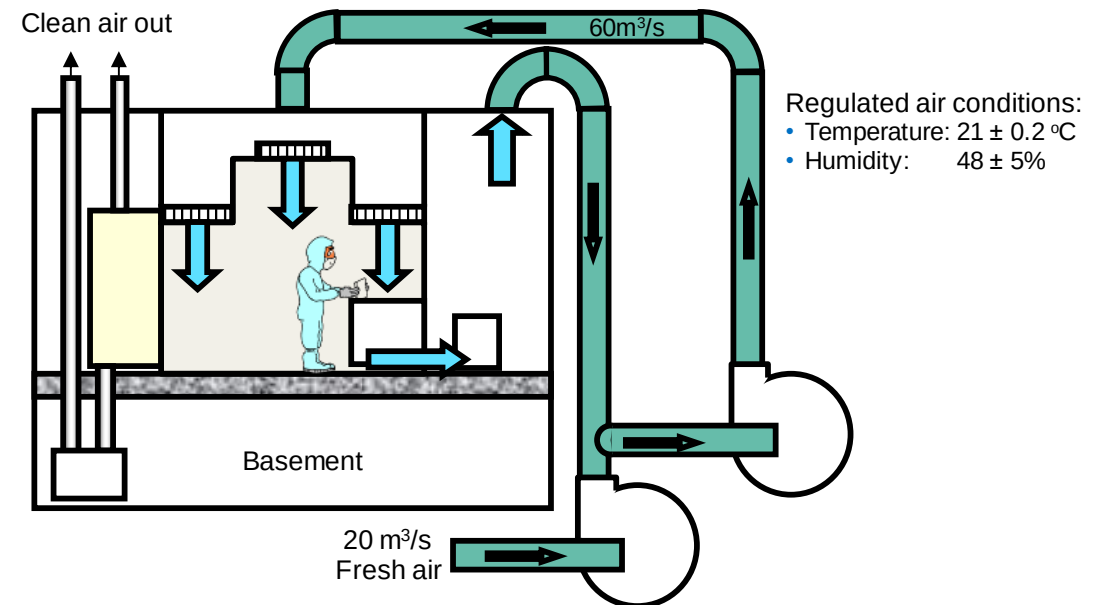
For bringing in visitors, contact: [Bruno Morana](#) or [Vincent van Croonenburg](#).

Cleanroom

A **clean room** is what your mother always wanted from you.



And a **cleanroom** is what we are discussing today.



Cleanroom definition

A cleanroom is defined as an enclosed area that is specially designed to control, monitor and maintain several parameters such as:

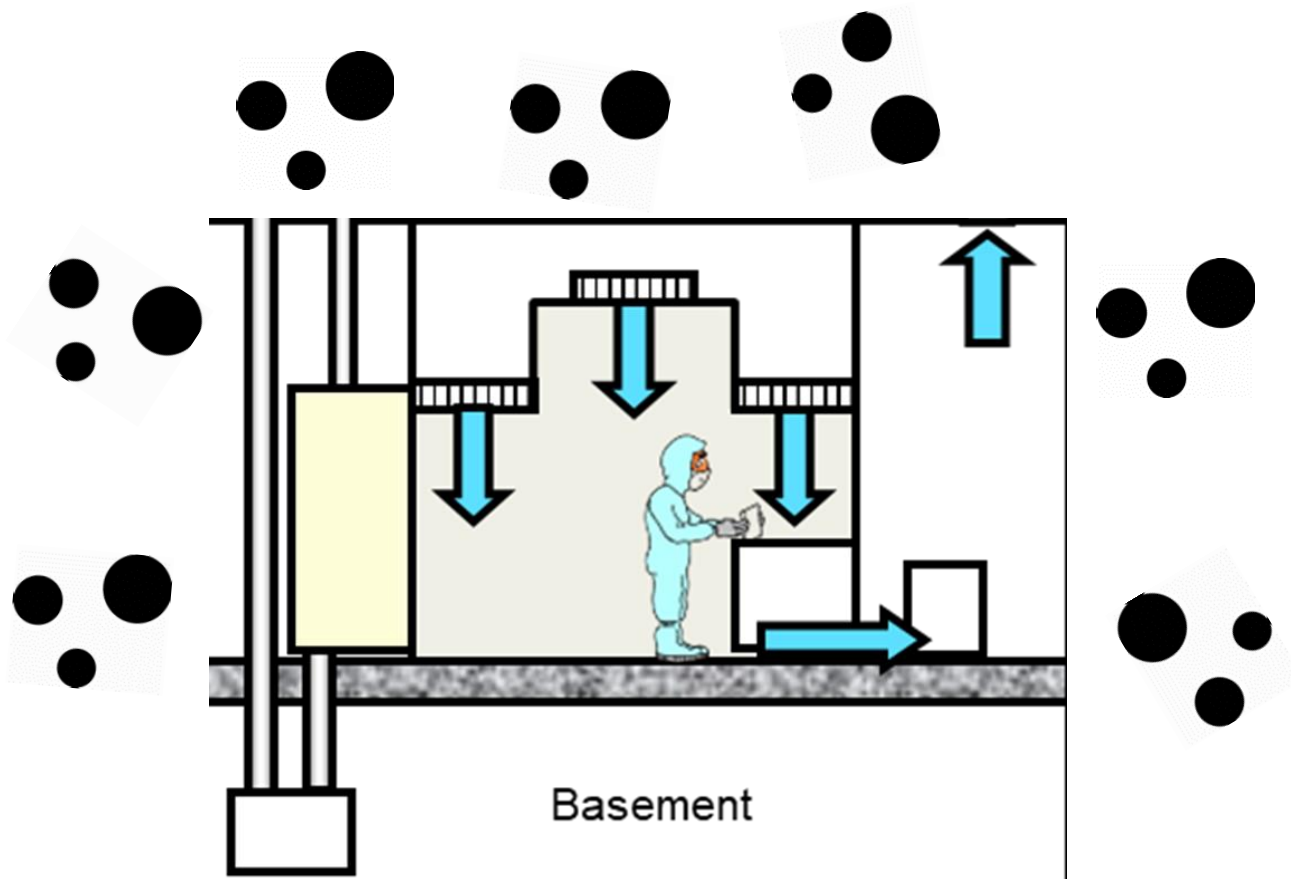
- air pollutant levels
- room pressure, temperature & humidity
- personnel access

to meet the critical environmental conditions that are demanded.

it's a

CONTROLLED ENVIRONMENT

Contamination classification



Two classification systems are in use:

1. Federal Standard 209E
2. ISO 14644-1

Cleanroom classification systems

2 cleanroom classification systems are in use to indicate the level of cleanliness. They show the maximum allowed number of particles of a certain size in a certain amount of air for each level.

EKL cleanrooms

US FED STD 209E Cleanroom Standards						ISO 14644-1
Class	Maximum particles/ft ³					ISO equivalent
	≥ 0.1 μm	≥ 0.2 μm	≥ 0.3 μm	≥ 0.5 μm	≥ 5 μm	
1	35	7.5	3	1	0.007	ISO 3
10	350	75	30	10	0.07	ISO 4
100	3,500	750	300	100	0.7	ISO 5
1,000	35,000	7,500	3,000	1,000	7	ISO 6
10,000	350,000	75,000	30,000	10,000	70	ISO 7
100,000	3.5 x 10 ⁶	750,000	300,000	100,000	700	ISO 8

EKL cleanrooms

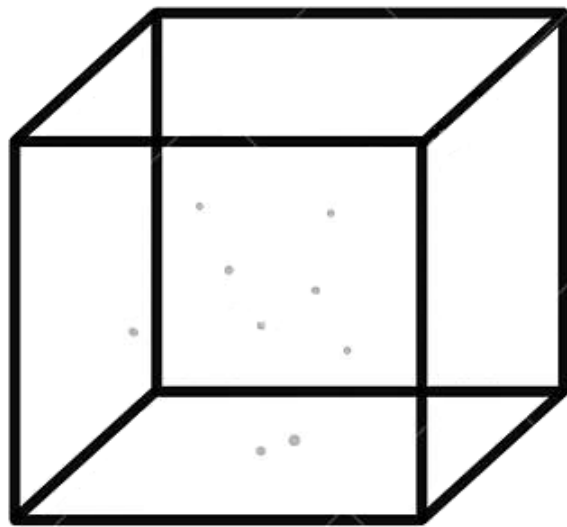


The numbers 1, 10, 100, 10000 denote the number of particles of size 0.5 μm or larger that are permitted per cubic foot of air.

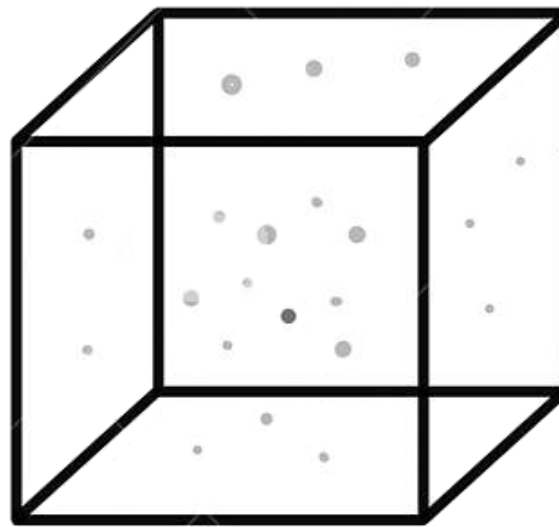
Class comparison

Number of Particles per Cubic Meter

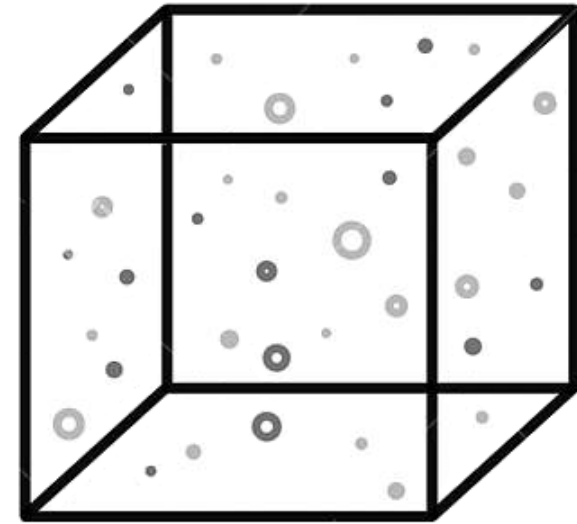
The level of cleanliness in the room by ISO 14644-1



Class ISO 5

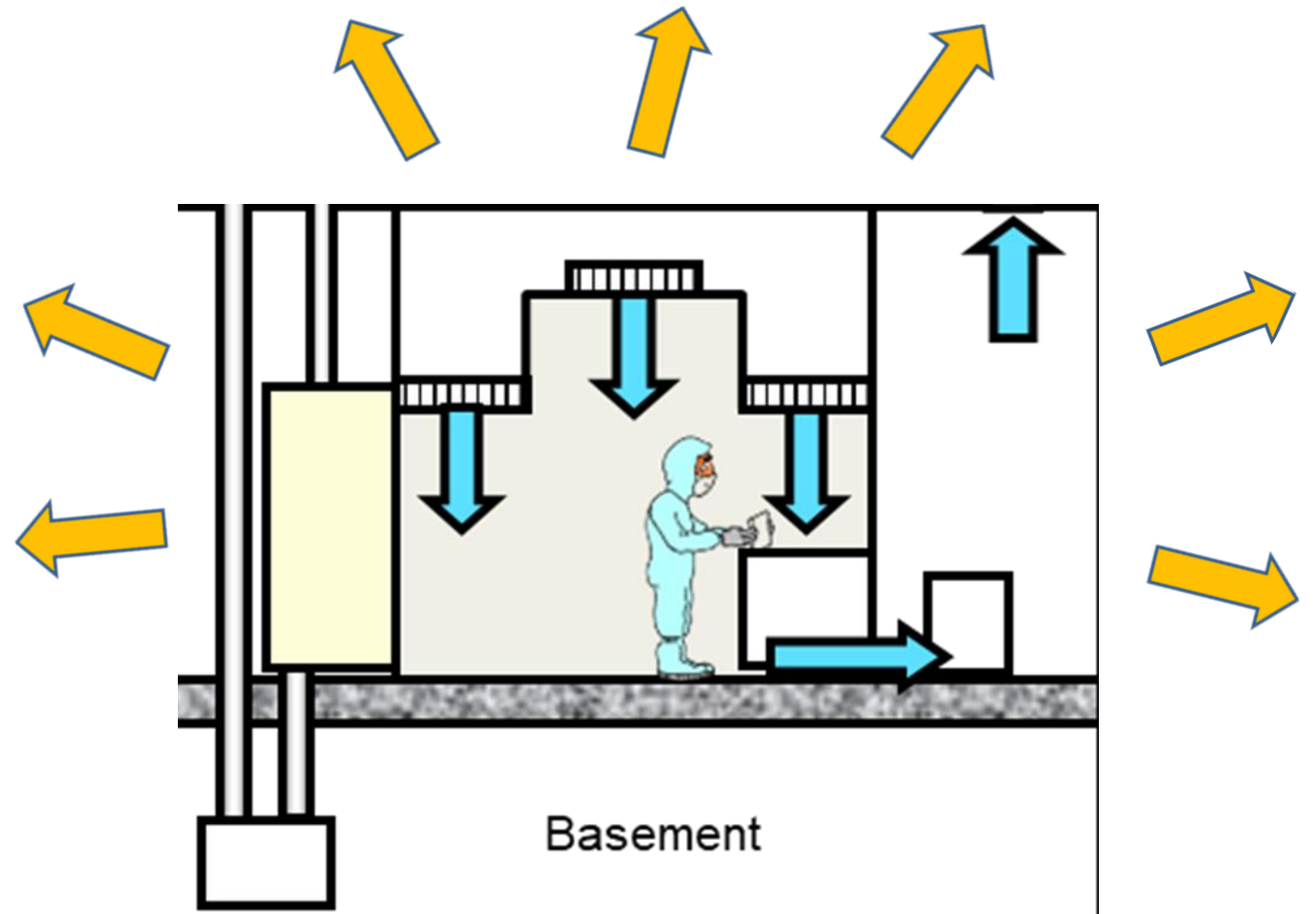
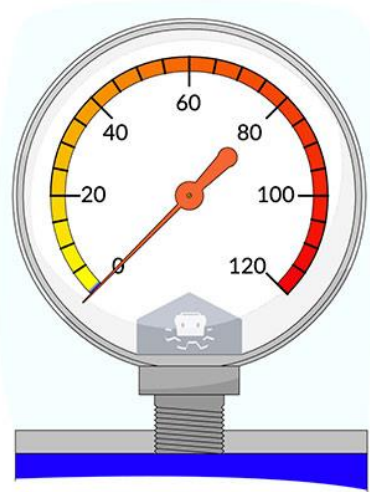


Class ISO 6



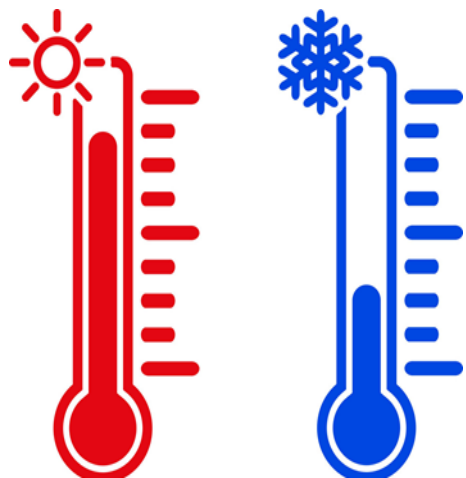
Class ISO 7

Pressurization



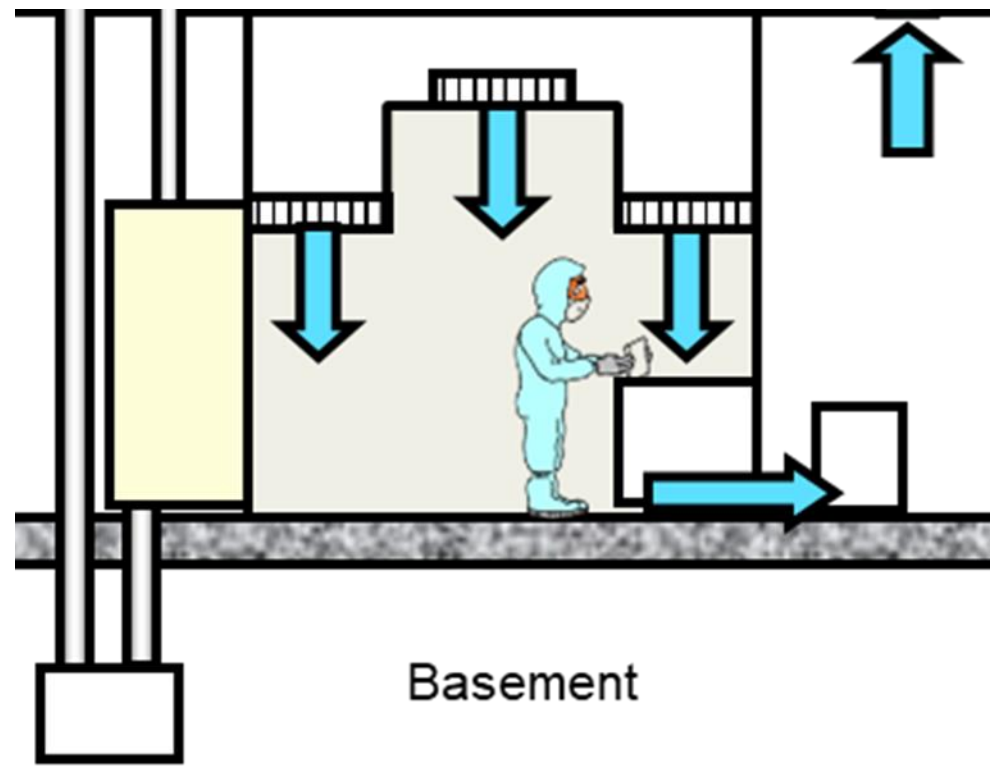
Pressurization is achieved by air pumps/fans, pumping air into the cleanroom through a HEPA filter to prevent dust from getting in. If anyone opens a door, air goes out instead of coming in from the outside.

Temperature and humidity

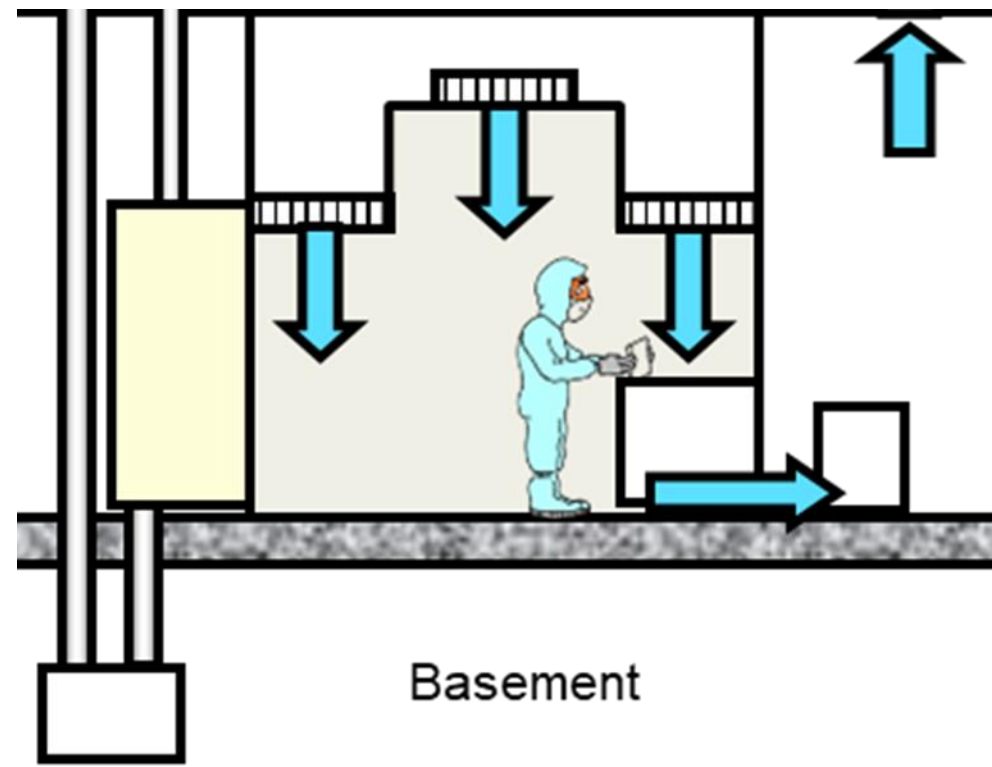
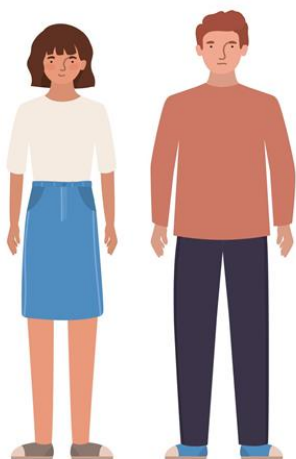


$T = 21 \pm 2 \text{ }^{\circ}\text{C}$

$\text{RH} = 48 \pm 2 \%$

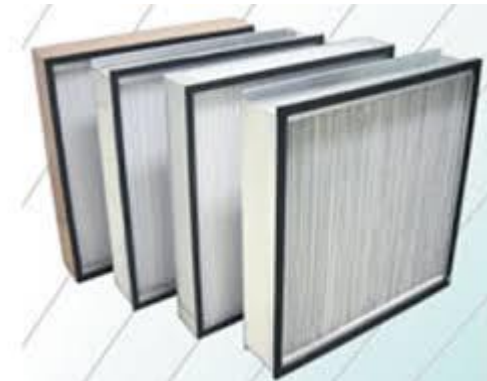


Personal access

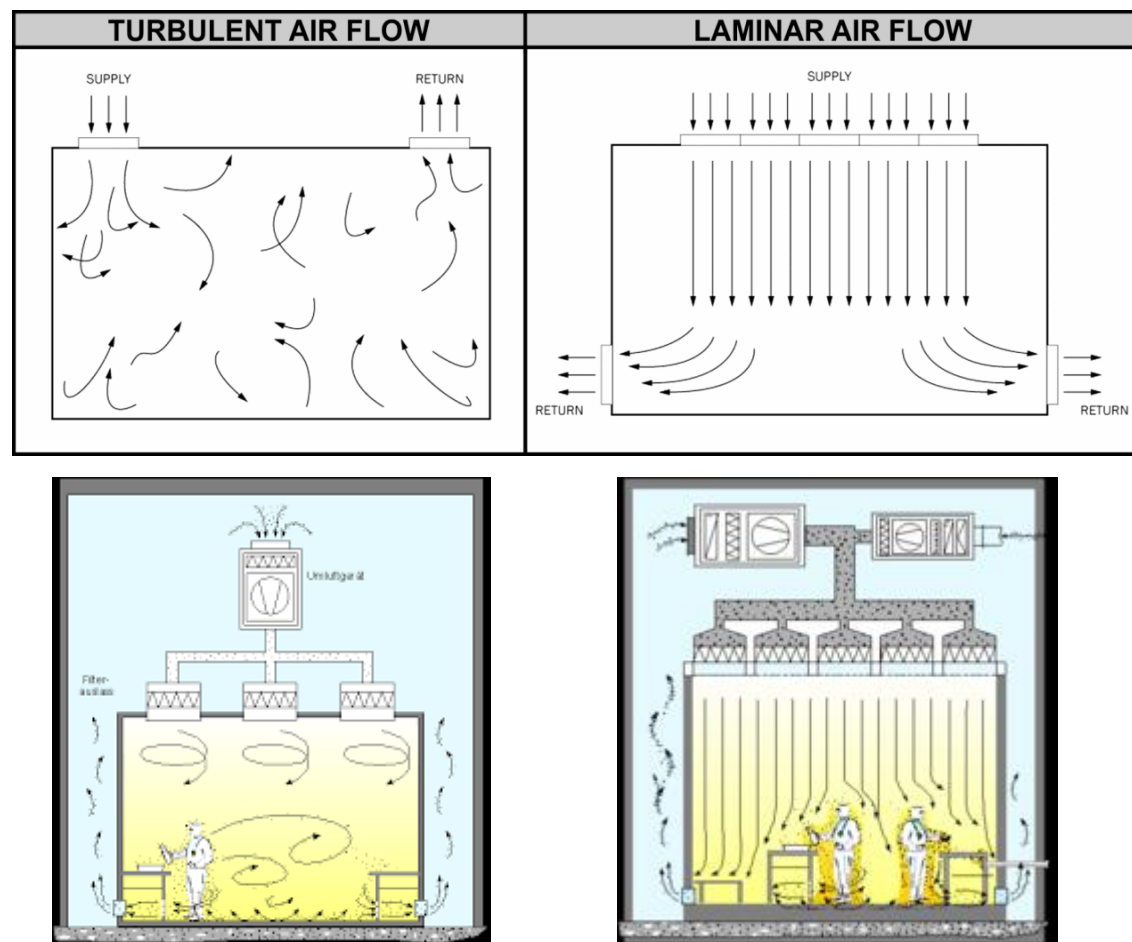


Principles of a clean environment

- Air is **filtered** (99.99% @ 0.3 μm) by a HEPA (High-Efficiency Particulate Air) filter.
- The **layout** should minimize particle sources in the filtered air stream.
- The **air flow** should remove most particles generated by processes.
(over-pressure / airflow)
- Selection of used **materials**.
- Applied **user protocols**.
- **Cleaning**.



Air flow



Prevent particles from swirling up

EKL infrastructure

Total laboratory space: 1570m²

Cleanroom ISO 5 (38%)

(Managed by EKL.)

Cleanroom ISO 7 (26%)

(Main user: PVMD group. Lab manager: [Martijn Tijssen](#).)

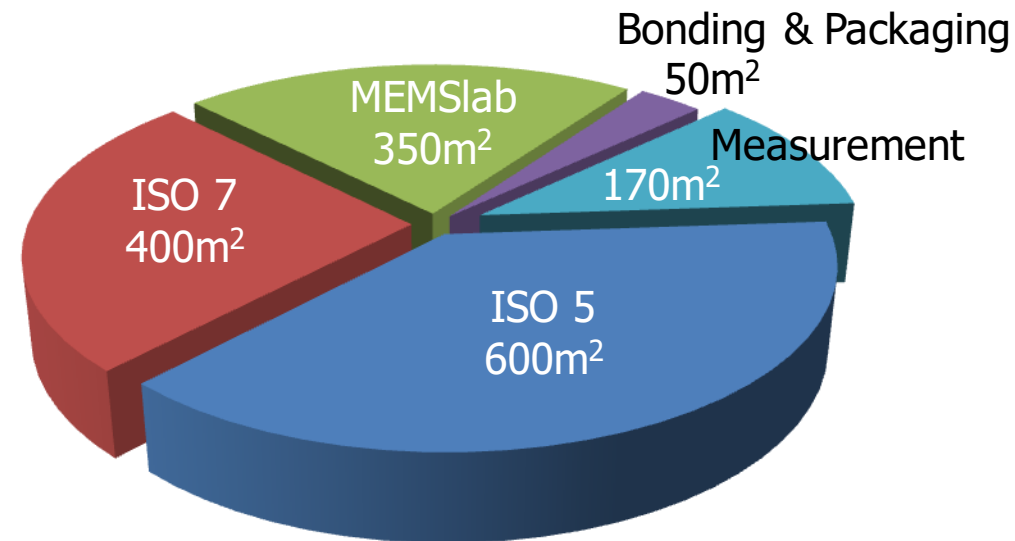
MEMS laboratory (22%)

(Managed by EKL.)

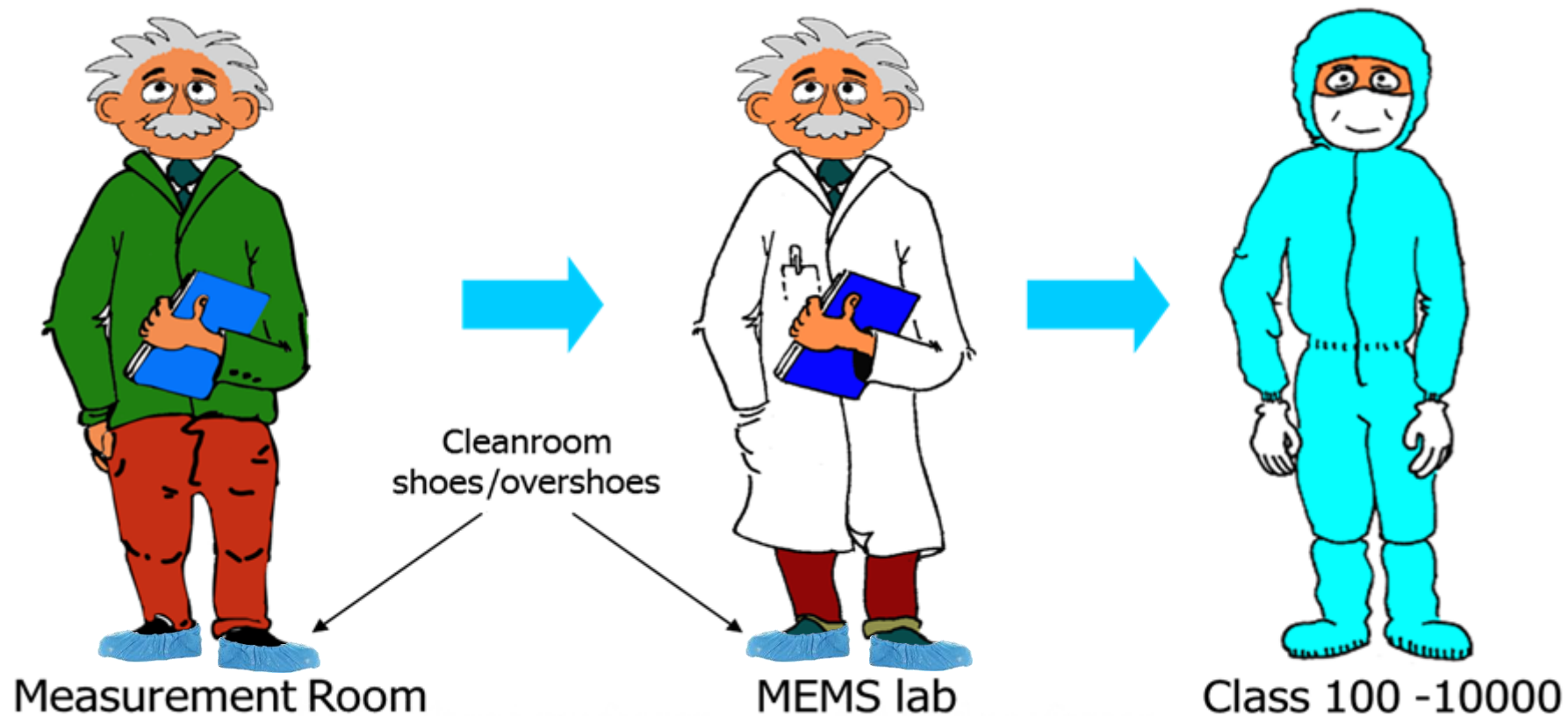
Electrical characterization laboratories (11%)

(Managed by EWI/ECTM. For access contact: [Filip Simjanoski](#).)

Bonding & Packaging laboratory (3%)



EKL lab clothing policy



Gowning / de-gowning procedure

- In the cleanroom all users must wear a **cleanroom suit** at all times. This suit includes: a coverall, a hood, a mouth cover, boots and white vinyl gloves.
- Please look at the EKL video film for the correct gowning procedure:
<https://www.youtube.com/watch?v=IJSUutm2wds>
- The instructions for wearing the cleanroom garments in order to prevent particle contamination is also posted on the **graphical guide** in the cleanroom gowning room.

User in training
(white hood)



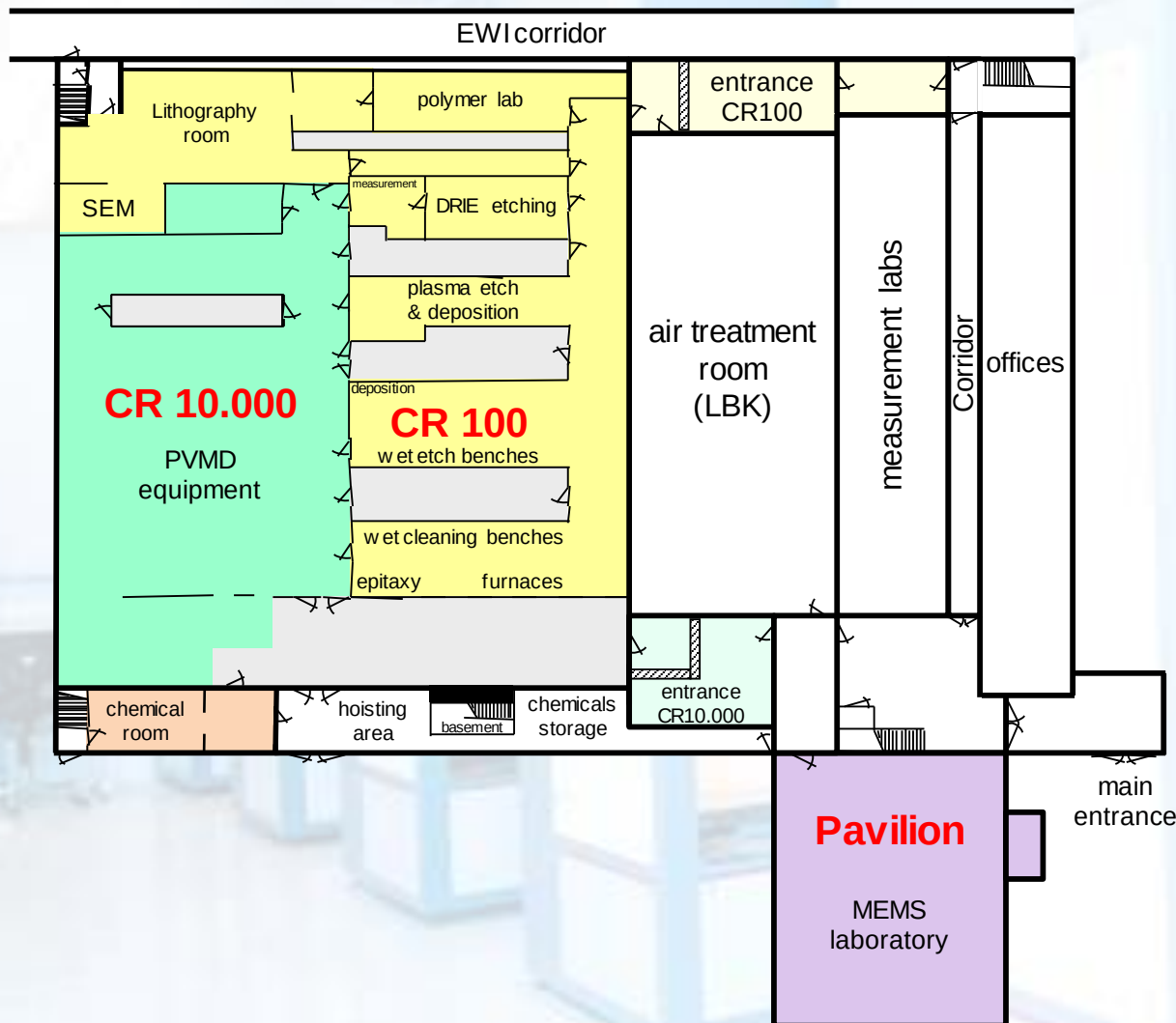
Trained user
(blue hood)



Service engineer
or cleaner



The Else Kooi Laboratories



- Cleanroom class 100
- Cleanroom class 10.000
- Service area (= class 10.000)
- Chemical room
- Pavilion (MEMS lab, bonding & packaging, ANL)

Some cleanroom rules

- Once you are dressed in a cleanroom suit and have entered the lab, you should **not unzip** the suit.
- If you need something from your pocket, you must return to the gowning area and retrieve it there.
- Items that are taken into any cleanroom must be wiped down with a **PurWipe microfiber** wipe. These wipes are moisturized with a water/IPA solution and can be found in the gowning areas.
- Generating particles in a cleanroom must be minimized at all time.
Hence it is **not allowed** to run, use normal paper, tear up paper, use a pencil, etc...
- It is allowed to take a mobile phone into the lab, but only when it is sealed in a plastic pouch that can be found in the gowning area.
- Eating and drinking in the labs is not allowed.

Equipment policy

- Access to the EKL labs **does not permit** the use of any equipment. Only certified persons are allowed to use the equipment in the labs unsupervised.
- Certification can only be obtained after a:
 - Process Integration Training (PIT)
 - Module training (MT)



Equipment use

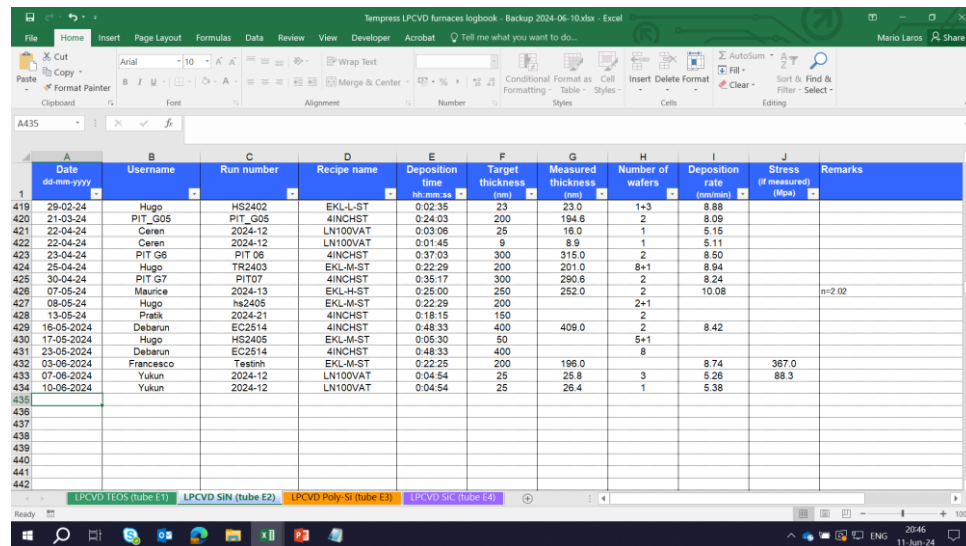
- All equipment in the facility (including chemical processing and inspection tools) has an **operating manual** available with the system and in NIS.
- The manuals describe the **procedures**, **specifications** and **limitations** of the equipment.
- The EKL cleanroom staff is responsible for the **development and revision** of equipment operating procedures.
- ANY use or operation of equipment that **falls outside** the procedure described in the operating manual must first be discussed with and approved by the tool owner.
- It is every user's responsibility to **properly operate** and **clean** each piece of equipment that they use.
- Equipment must be left behind in the **default IDLE state**.
(If you find a system in an incorrect state, then leave it in the correct state after use.)

Equipment reservation etiquette

- Equipment reservations can be made in the [NanoLabNL Information System](#) (NIS).
- Reservations are given a [15 minute late grace](#) period on the starting time.
- Users without reservations must yield the equipment to anyone with a proper reservation.
- Users should delete reservations that they cannot fulfill as soon as possible.
- [Keep reservations as accurate as possible](#). Do not reserve a 4-hour time period for 1 hour of process work.
- Please delete excess reservation time when you're ready, so others can use the system.
- If you need to reserve equipment for an extended period ([>6 hours](#)), then please use time which is outside the standard hours. Start your process after 5 PM and let it run overnight.

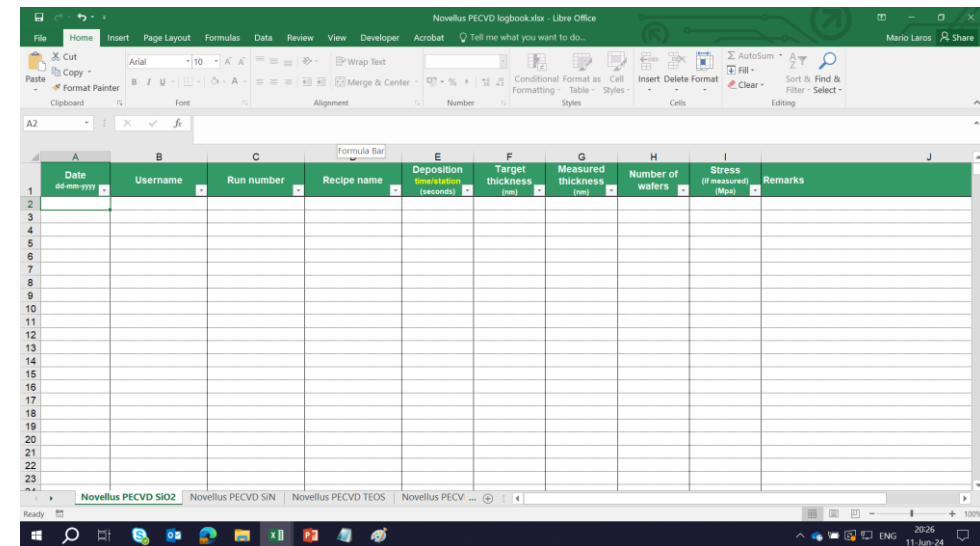
Equipment log book

- All users of equipment **must** fill in the **equipment log books (excel files)**, and record all process runs and results.
- Recent logbook results from previous users can be used to calculate process times.



Date	Username	Run number	Recipe name	Deposition time	Target thickness	Measured thickness	Number of wafers	Deposition rate	Stress	Remarks
29-02-24	Hugo	HS2402	EKL-L-ST	0:02:35	23	23.0	1+3	8.88		
21-03-24	PIT_G05	PIT_G05	4INCHST	0:24:03	200	184.6	2	8.09		
22-04-24	Ceren	2024-12	LN100VAT	0:03:06	25	16.0	1	5.15		
23-04-24	Ceren	2024-12	LN100VAT	0:01:45	9	8.9	1	5.11		
23-04-24	PIT_G6	PIT_G6	4INCHST	0:37:03	300	315.0	2	8.50		
25-04-24	Hugo	TR2403	EKL-M-ST	0:22:29	200	201.0	8+1	8.94		
30-04-24	PIT_G7	PIT_G7	4INCHST	0:35:17	300	290.6	2	8.24		
07-05-24	Meurice	2024-13	EKL-H-ST	0:25:00	250	252.0	2	10.08		
08-05-24	Hugo	hs2405	EKL-M-ST	0:22:29	200		2+1			n=2.02
13-05-24	Pratik	2024-21	4INCHST	0:18:15	150		2			
16-05-2024	Debanun	EC2514	4INCHST	0:48:33	400	409.0	2	8.42		
17-05-2024	Hugo	HS2405	EKL-M-ST	0:05:30	50		5+1			
23-05-2024	Debanun	EC2514	4INCHST	0:48:33	400		8			
03-06-2024	Francesco	Testinh	EKL-M-ST	0:22:25	200	196.0		8.74	367.0	
07-06-2024	Yukun	2024-12	LN100VAT	0:04:54	25	25.8	3	5.26	88.3	
10-06-2024	Yukun	2024-12	LN100VAT	0:04:54	25	26.4	1	5.38		

Excel logbook file at LPCVD furnaces.

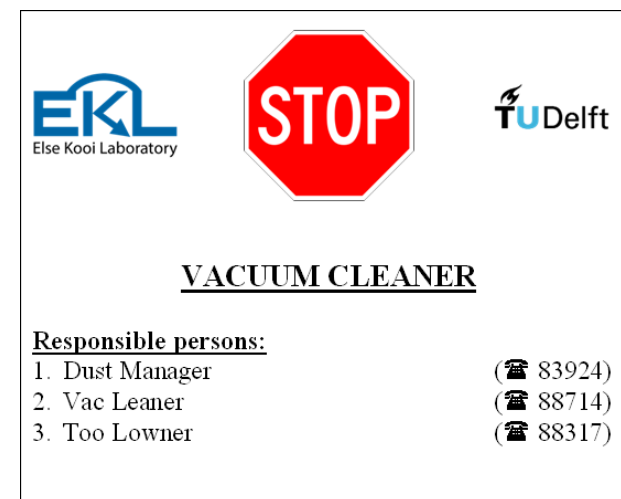


Date	Username	Run number	Recipe name	Deposition time	Target thickness	Measured thickness	Number of wafers	Stress	Remarks
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Libre Office logbook file at Novellus PECVD system.

Equipment status

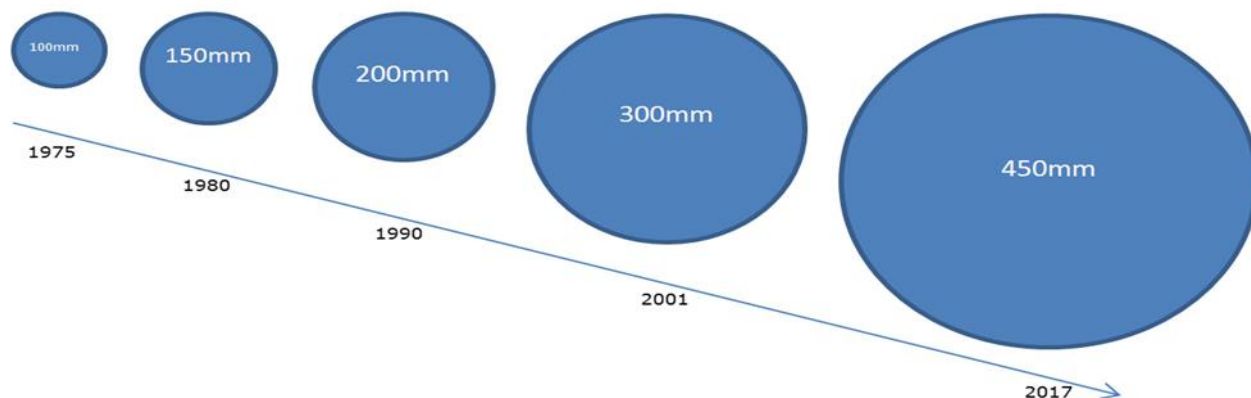
- Equipment abnormalities, problems with operation of a machine, deviations in process quality and broken wafers must be reported to the staff member(s) responsible for the machine.
- The first step is to attach a **STOP** sign on the front of the tool.
- The second step is to change the machine status in the [NanoLabNL Information system \(NIS\)](#). (An [e-mail](#) will automatically be sent to the responsible staff members.)
- A list of tool owners with their telephone number is posted on every tool and can be found in NIS.
- It is forbidden to use equipment which is reported down in NIS, or labeled with a stop sign.
- Do not repair or take corrective actions on any tool by yourself.



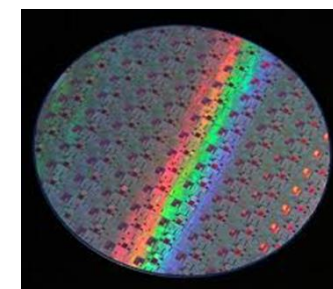
Equipment training - Golden rules

- Advance planning is very **important**: Max. response time for a training appointment is 2 **days**. A module training is to be performed within 1 **week**.
- Read the equipment or procedure manual before the start of your training session.
- You are encouraged to bring your own **samples**.
- After attending a training session you are allowed to **operate a machine** with the help of your colleagues, who have been authorized to operate the machine independently (= allowed user).
- Once the trainer is fully satisfied with your knowledge and ability, you will be authorized and are put on the **allowed users** list in NIS.

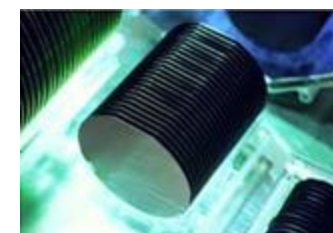
Available wafer types



single crystal Silicon ingot



Si wafer with microchips



Standard types Silicon wafers at EKL:

100 mm diameter (4 inch) ➡ p and n type

150 mm diameter (6 inch) ➡ very limited

Different types of glass wafers (100 mm) are also present, but only in limited amounts.

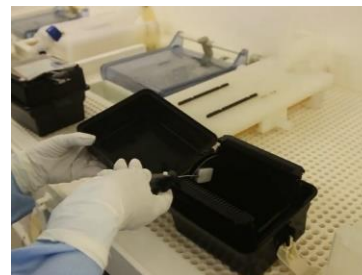
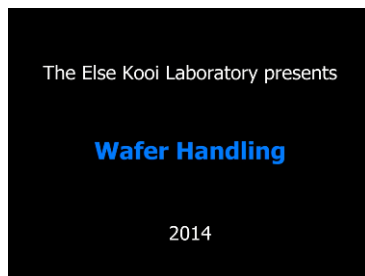
(Some types contain contaminating materials which restricts their use.)

Wafer handling

- All wafers should be properly **covered** at all times.
⇒ Keep them in a wafer box and do not leave the box open unnecessarily.
- **Never touch** wafers with gloved or ungloved hands.
- Only use clean **vacuum** or **manual tweezers** to handle a wafer from the **backside** or **edge**.
- It is **not allowed** to use vacuum tweezers on contaminated wafers.
- Do not use manual tweezers with a **metal beak**. This may contaminate your wafer with iron (Fe).
- Label manual tweezers that are used for **contaminated** wafers.
⇒ To prevent cross-contamination it is **not allowed to use them for non-contaminated wafers** anymore.
- Check that a vacuum tweezer is working properly before attempting to transfer a wafer.
- Clean a vacuum tweezer if needed with IPA before handling a wafer. (Use a **PurWipe microfiber** wipe.)
- **Never hold your hand or face above a wafer or wafer carrier and do not grab a carrier from the inside.**

Wafer box handling

- Any time that wafers are to be transported, they must be stored in a box that is **designed** for the wafers.
- Always use **both hands** when moving wafer boxes.
- Never hold the box against the body.
- While wafers are in a box, never put the box **on tools**.
- **For the proper wafer handling procedure, check out the EKL video at:**
https://www.youtube.com/watch?v=X55D05V_aRU



Ethics & Good laboratory practice

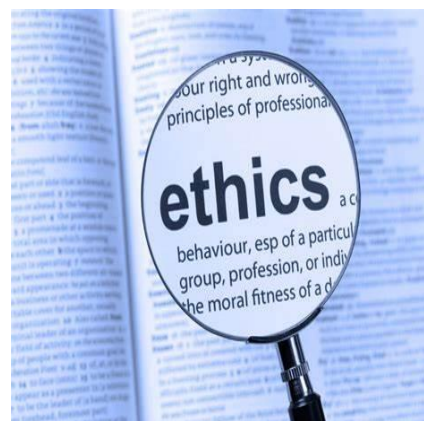
The success of each Lab User relies on:

- Understanding
- Shared responsibility among all users
- Participation
- Self discipline



Some behaviour rules

- If you **OPEN** it, **CLOSE** it!
- If you **TURN** it **ON**, **TURN** it **OFF**!
- If you **UNLOCK** it, **LOCK** it!
- If you can't **FIX** it, **CALL IN SOMEONE WHO CAN**!
- If you **BORROW** it, **RETURN** it!
- If you **USE** it, **TAKE CARE** of it!



- If you make a **MESS**, **CLEAN** it **UP**!
- If you **MOVE** it, **PUT** it **BACK**!
- If it belongs to **SOMEONE ELSE**, ask **PERMISSION** to use it!
- If you **DON'T KNOW** how to operate it, **LEAVE IT ALONE**!
- If it doesn't **CONCERN YOU**, **DON'T MESS WITH IT**!

House keeping rules

- Every user is responsible for keeping the labs clean.
- Label your wafer boxes and store them in your cupboard.
- Store your notebook, tweezers, glasses and timer in your cupboard.
- Always and correctly label containers with chemicals.
- Always clean your workspace before leaving.
- Properly store all used materials before leaving the cleanroom.
- Don't leave or store items in or on equipment.
- The **last person** leaving the cleanroom should make sure that it's in a safe idle mode.



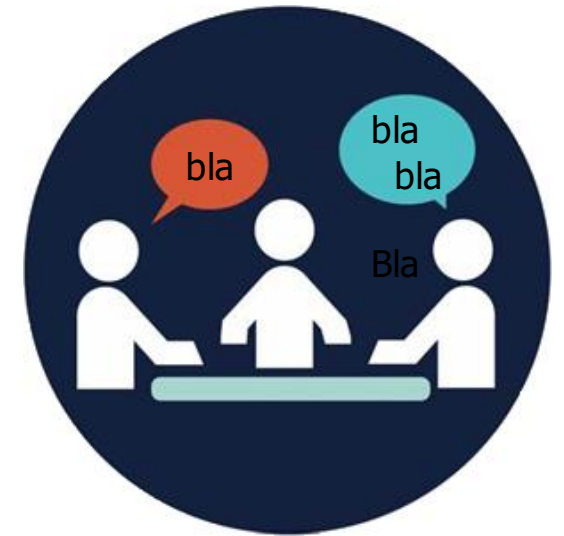
Lab user meeting

An interactive lab user meeting is organized on a regular basis to:

- exchange information
- ask questions
- share complaints
- discuss issues
- and

about lab equipment, processes, facility maintenance, lab closures, etc...

An invitation to join this meeting on **Microsoft Teams** will be sent by the EKL Lab Manager (Bruno Morana).



EKL's rule violation protocol

- First time violation of rules:

- Verbal warning.
- Possibly a "White Box" punishment.



- Second or moderate violation:

- Talk with the Safety Officer
Re-training may be required.

- Third time or severe violation:

- Cleanroom access will be withdrawn.
- Access will not be renewed, until an acceptable resolution is reached.
- Cleanroom re-training will be required.



